INCH-POUND

MIL-E-44231B
4 April 1990
SUPERSEDING
MIL-E-44231A
14 March 1988

MILITARY SPECIFICATION

ESCALLOPED POTATOES, THERMOSTABILIZED, TRAY PACK

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers escalloped potatoes, thermostabilized in tray pack cans for use by the Department of Defense as a component of operational rations.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

* 2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.1).

SPECIFICATIONS

FEDERAL

PPP-B-636 - Boxes, Shipping, Fiberboard

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8940

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MILITARY

MIL-L-1497 - Labeling of Metal Cans for Subsistence Items
MIL-L-35078 - Loads, Unit; Preparation of Semiperishable Subsistence Items; Clothing, Personal Equipment and Equipage; General Specification For
MIL-C-44340 - Can, Tray Pack

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-900 - Bacterial Standards for Starches, Flours, Cereals, Alimentary Pastes, Dry Milks and Sugars Used in the Preparation of Thermostabilized Foods for the Armed Forces

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

* 2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

ENVIRONMENTAL PROTECTION AGENCY (EPA)

National Primary Drinking Water Regulations

(Copies are available from the Office of Drinking Water, Environmental Protection Agency, WH550D, 401 M Street, S.W., Washington, DC 20460.)

U.S. DEPARTMENT OF AGRICULTURE (USDA)

Meat and Poultry Inspection Regulations

(Copies area available from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402-0001.)

U.S. Standards for Condition of Food Containers

(Copies are available from the Chairman, Condition of Container Committee, Agricultural Marketing Service, Room 2506, South Building, P.O. Box 96456, U.S. Department of Agriculture, Washington, DC 20090-6456.)

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U.S. DEPARTMENT OF HEALTH AND HUMAN SERVICES (HHS)

Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder (21 CFR Parts 1-199)

(Copies are available from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402-0001.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.1).

AMERICAN ASSOCIATION OF CEREAL CHEMISTS (AACC)

Approved Methods of the American Association of Cereal Chemists

(Application for copies should be addressed to the American Association of Cereal Chemists, 3340 Pilot Knob Road, St. Paul, MN 55121.)

AMERICAN DEHYDRATED ONION AND GARLIC ASSOCIATION (ADOGA)

Official Standards and Methods of the American Dehydrated Onion and Garlic Association for Dehydrated Onion and Garlic Products

(Application for copies should be addressed to the American Dehydrated Onion and Garlic Association, 650 California Street, Suite 800, San Francisco, CA 94108.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3330 - Peel Adhesion of Pressure-sensitive Tape

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

ASSOCIATION OF OFFICIAL ANALYTICAL CHEMISTS (AOAC)

Official Methods of Analysis of the Association of Official Analytical Chemists

(Application for copies should be addressed to the Association of Official Analytical Chemists, 2200 Wilson Boulevard, Suite 400-CD, Arlington, VA 22201-3301.)

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NATIONAL ACADEMY OF SCIENCES

Food Chemicals Codex

(Application for copies should be addressed to the National Academy Press, 2101 Constitution Avenue, N.W., Washington, DC 20418.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.1), a sample shall be subjected to first article inspection (see 6.2) in accordance with 4.4.

3.2 Ingredients. All ingredients shall be clean, sound, wholesome, and free from foreign material, evidence of rodent and insect infestation, extraneous material, off-flavors, off-odors, and off-colors.

3.2.1 Potatoes, fresh. Potatoes shall be fresh, firm and of a white flesh variety suitable for canning. The maximum specific gravity for the potatoes shall be 1.075 with a reducing sugar content of not more than 2.0 percent on a dry weight basis.

3.2.2 Water. Water used for ice making, formulation, and washing shall conform to the National Primary Drinking Water Regulations.

3.2.3 Shortening, vegetable, powdered. The powdered vegetable shortening shall consist of components which, when combined, will serve as a product whitening agent. The powdered shortening, when used as an ingredient in this product, shall withstand the conditions of thermoprocessing specified in this document without browning. A typical analysis of this powdered shortening follows:

Fat	-	75.0 percent ± 1.5 percent
Protein	-	5.5 percent ± 0.5 percent
Carbohydrates	-	15.0 percent ± 1.0 percent
Moisture	-	2.0 percent maximum

3.2.4 Starch, modified, high opacity. The high opacity starch shall be white, odorless, finely pulverized modified maize food starch for use in canned foods. The modified starch shall demonstrate initial viscosity development in the temperature range of 140° to 170°F and a typical viscosity (be fully hydrated) at common retort temperatures. The starch shall resist breakdown at

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low pH, under shear stress, and under conditions of cold storage. The starch shall be bland with essentially no cereal or starch taste and shall comply with MIL-STD-900.

* 3.2.5 Margarine. Margarine shall conform to the FDA Standards of Identity for Margarine (21 CFR 166.110) and shall possess a fine and pleasing flavor. The body shall be smooth, firm, and homogeneous. The margarine shall possess a uniform medium yellow color and may be salted or unsalted.

3.2.6 Salt. Salt shall be noniodized, white, refined sodium chloride with or without anticaking agents.

3.2.7 Broth, chicken. The chicken broth may be purchased frozen or canned (thermostabilized). The broth shall be clear, essentially fat free, with percent solids sufficient to comply with the solids requirements in the sauce formulation. The chicken broth shall have a characteristic mild chicken broth flavor and odor and may contain flavor enhancers approved by the Food and Drug Administration. The chicken broth shall be produced in a USDA inspected plant in accordance with USDA regulations and when frozen shall have been held at 0°F or below for not more than 75 days prior to sauce formulation.

3.2.8 Onion flavoring. Onion flavoring shall possess a characteristic onion odor and flavor, shall contain little or no fruit sugar or dextrose, and shall be carried on noniodized salt with or without anticaking agents.

3.2.9 Lecithin. Lecithin shall comply with the Food Chemicals Codex.

3.2.10 Onion powder. Onion powder shall be fancy grade of the Official Standards and Methods of the American Dehydrated Onion and Garlic Association for Dehydrated Onion and Garlic Products.

3.2.11 Pepper, white, ground. Ground white pepper shall be derived from the mature berries of Piper nigrum L., from which the outer covering or outer and inner coverings have been removed. The ground pepper shall have a characteristic penetrating odor, a hot biting pungent flavor, and a light color. Volatile oil content shall be not less than 1.0 mL/100 grams, and the white pepper shall be of such size that not less than 95 percent shall pass through a U.S. Standard No. 40 sieve.

3.2.12 Celery flavoring, soluble. Soluble celery flavoring shall be prepared by properly blending and dispersing a minimum of 4 percent by weight of total dried fruit extractives of Apium graveolens L. on a soluble, dry, edible carrier. The flavoring shall possess a clean, celery-like aroma with a slightly bitter taste, and shall be yellowish gray in color, free flowing, and free from caking and all extraneous matter.

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3.3 Preparation and processing. Processing shall be on a continuous basis.

3.3.1 Potato preparation. The potatoes shall be cleaned to remove dirt and extraneous material, peeled, washed thoroughly, trimmed, and cut into approximately 1/4 ± 1/16 inch slices. The potato slices shall be blanched sufficiently to prevent discoloration and to remove excess air. The blanched potatoes shall be immediately cooled to the initial temperature of the cooling water and thoroughly drained. The cooled drained potatoes shall be handled in a manner to prevent discoloration and shall be filled into the tray pack cans within 4 hours after blanching.

* 3.3.2 Sauce preparation. The sauce shall be formulated and prepared as follows:

Ingredients	Percent by weight
Water	86.67
Powdered shortening	4.00
Starch, modified	4.00
Margarine	2.00
Salt 1/	1.25
Chicken broth (15 to 18 percent solids) 2/	1.00
Onion flavoring	0.50
Lecithin	0.25
Onion powder	0.25
Pepper, white, ground	0.07
Celery flavoring	0.01

1/ The total amount of salt in the sauce shall be adjusted as necessary to produce a product that complies with the finished product salt requirements (see 3.6.m).

2/ Chicken broth with different solids content shall be adjusted to 15 to 18 percent.

NOTE: The following sauce preparation procedures were used in the development of this product. Alternative procedures may be used provided finished product requirements are met. (When alternative procedures are used, the time and temperature requirements specified for the prepared sauce are still applicable.)

The starch and powdered shortening shall be mixed with sufficient water to obtain a thin, smooth slurry. The remaining ingredients shall be mixed in a steam-jacketed kettle and heated to 180° to 190°F. The starch slurry shall be added with vigorous mixing to obtain a smooth, lump-free mixture. The entire contents shall be heated to 185°F with continuous agitation and held for 5 minutes. The volume of the final mixture shall be adjusted with water before tray filling to compensate for evaporation loss during heating. The sauce shall be maintained in the temperature of 150° to 180°F and filled into tray pack cans within 4 hours after preparation.

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3.4 Tray pack filling and sealing. Each tray pack can (see 5.1.1) shall be filled with layered potato slices and sauce such as to conform to the finished product requirements and to the following requirements.

a. Immediately after filling each can shall be hermetically sealed under a vacuum established by a processing authority and specified in the scheduled process so as to ensure compliance with the finished product requirement (see 3.6.n).

b. The filled and sealed tray pack cans shall be in the retort process within 2 hours after sealing.

3.5 Tray pack thermoprocessing. The filled and sealed tray pack cans shall be thermostabilized by retorting until a sterilization value (F_o) of not less than 6.0 has been achieved. The filled and sealed thermoprocessed tray pack cans shall show no evidence of can swelling when tested for commercial sterility as specified in 4.5.3.4.

3.6 Finished product requirements. The finished product shall comply with the following requirements:

a. There shall be no foreign material such as, but not limited to, dirt, insect parts, hair, wood, glass, or metal.

b. There shall be no foreign odor or flavor such as, but not limited to, burnt, scorched, stale, sour, rancid, or moldy.

c. There shall be no color foreign to the product.

d. No individual can shall contain less than 104 ounces of product.

e. The average net weight shall be not less than 106 ounces.

f. The average drained weight of potato slices shall be not less than 66.0 ounces.

g. No individual can shall contain less than 64.0 ounces drained weight of potato slices.

h. Potato texture shall be firm and not mushy, hard, or fibrous.

i. No less than 85 percent (by weight) of the potatoes shall be whole intact slices.

j. The diameter of the potato slices shall be governed by the diameter of the potatoes from which they have been derived, but at least 50 percent (by weight) of the slices shall be in the range of 1.5 to 2.5 inches in diameter.

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k. The sauce shall be smooth, opaque, moderately thick, and creamy off-white in color.

l. The product shall show no evidence of excessive heating (materially darkened, scorched).

m. The salt content of any individual tray pack shall be not less than 0.5 percent nor greater than 1.3 percent.

n. Filled, sealed, and retorted cans shall show evidence of proper vacuum as determined by concavity of the can lid (see 4.5.6).

3.6.1 Palatability. The product shall be equal to or better than the approved preproduction sample (see 6.1) in palatability and overall appearance.

3.7 Plant qualification. The product shall be prepared, processed, and packaged in establishments meeting the requirements of 21 CFR, Part 110, "Current Good Manufacturing Practice in Manufacturing, Processing, Packaging or Holding Human Food," and the plant sanitation requirements of the appropriate Government inspection agency.

3.8 Federal Food, Drug, and Cosmetic Act. All deliveries shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor's responsibility. Inspection and acceptance by the USDA shall not relieve the contractor of obligation and responsibility to deliver a product complying with all requirements of this specification. The contractor shall ensure product compliance prior to submitting the product to the USDA for any inspection.

4.2 Inspection and certification. Product acceptability shall be determined by the USDA. The USDA will determine the degree of inspection and supervision necessary to ensure compliance with the requirements of this specification.

4.3 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).
b. Quality conformance inspection (see 4.5).

4.4 First article inspection. When a first article is required (see 6.1), it shall be inspected in accordance with the quality assurance provisions of this specification and evaluated for overall appearance and palatability. Any failure to conform to the quality assurance provisions of this specification or any appearance or palatability failure shall be cause for rejection of the first article.

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* 4.5 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.5.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

* 4.5.1.1 Ingredient and component examination. Conformance of ingredients and components to identity, condition, and other requirements specified in 3.2 shall be certified by the ingredient supplier or ingredient manufacturer, and compliance shall be verified by examination of pertinent labels, markings, U.S. Grade certificates, certificates of analyses, or other such valid documents acceptable to the inspection agency. If necessary, each ingredient shall be examined organoleptically or inspected according to generally recognized test methods, such as the standard methods described in the Official Methods of Analysis of the Association of Official Analytical Chemists and in the Approved Methods of the American Association of Cereal Chemists, to determine conformance to the requirements. Any nonconformance to an identity, condition, or other requirement shall be cause for rejection of the ingredient or component lot or of any involved product.

4.5.2 In-process examination. In-process examination shall be performed to determine conformance to the preparation, processing, can interior coating, filling, sealing, and packaging requirements. Any nonconformance revealed by actual examination or by review of records of time, temperature, and formulation or of other valid documents shall be cause for rejection of the involved product.

4.5.3 Tray pack inspection. The inspection lot shall include only tray packs produced in one workshift. The USDA reserves the right to separate the inspection lot into smaller inspection lots.

4.5.3.1 Net weight inspection. Randomly select 30 filled and sealed tray pack cans from the inspection lot and weigh separately. Subtract the average tare weight (determined by randomly selecting and weighing 30 of the empty tray pack cans and lids used in preparing the product and dividing the total weight by 30) from the weight of each tray pack in the sample. The results shall be reported to the nearest 1 ounce. If the average net weight is less than 106 ounces or if the net weight of any individual can is less than 104 ounces, the lot shall be rejected.

* 4.5.3.2 Product inspection. The sample size shall be as indicated by the double sampling plan specified in table I. The sample cans shall be selected at random from the lot. The sample cans shall be heated in accordance with heating instructions on the can label, opened, and inspected for the defects listed in table II.

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TABLE I. Double sampling plan for product inspection 1/

Lot size (cans)	Sample size (cans)	Cumulative sample	Acceptance number	Rejection number
0 to 3200	8	-	0	2
	8	16	1	2
3201 to 35000	13	-	0	3
	13	26	3	4

1/ a. If no defects are found in the first sample, the lot shall be accepted.

b. If the number of defects found in the first sample equals or exceeds the rejection number, the lot shall be rejected.

c. If the number of defects found in the first sample exceeds the acceptance number but is less than the rejection number, the second sample shall be inspected. Defects found in the first and second samples shall be combined and if the number of defects in the cumulative sample equals or exceeds the rejection number, the lot shall be rejected.

TABLE II. Product defects 1/ 2/ 3/

Category	Defect
Major	
101	Drained weight of potato slices in a can is less than 64.0 ounces 4/ 5/
102	Texture of potato slices hard, fibrous, or mushy
103	Less than 85 percent, by weight, whole intact potato slices
104	More than 50 percent, by weight, of potato slices measuring less than 1.5 inches or more than 2.5 inches in diameter

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TABLE II. Product defects 1/ 2/ 3/ - continued

Category	Defect
Major	
105	Sauce consistency not smooth, opaque, and moderately thick
106	Sauce not creamy off-white in color
107	Product shows evidence of excessive heating (materially darkened, scorched)

1/ The presence of foreign material (for example, dirt, insect parts, hair, wood, glass, metal), foreign odor or flavor (for example, burnt, scorched, moldy, rancid, sour, stale), or foreign color shall be cause for rejection of the lot.

2/ Product not equal to or better than the approved preproduction sample in palatability or overall appearance shall be cause for rejection of the lot. (This comparison shall be performed only when deemed necessary by an Agricultural Marketing Service (AMS) agent.)

3/ Interior of filled and sealed cans shall be examined visually for coating defects during product examination. Suspected rust spots shall be verified by testing in accordance with 4.5.7. Any spot verified as a rust spot shall be cause for rejection of the lot. Any scratch or fracture that penetrates through the coating of a can shall be cause for rejection of the lot.

4/ To determine drained weight, the free liquid in the can shall be poured off and the remaining contents shall be poured into a U.S. Standard 1/4-inch sieve in a manner that will distribute the product over the sieve without breaking the potato slices. Sieve area shall be such that the distributed product does not completely cover all the openings of the sieve. The sieve and contents shall be immersed in 140° to 160°F water, and the sieve agitated with a vigorous swirling motion for 1 minute taking care to separate the product into individual slices. The sieve and contents shall be removed from the water and then reimmersed momentarily twice in succession. The sieve shall be tilted at approximately a 45-degree angle and allowed to drain for 2 minutes before determining the drained weight by subtracting the sieve tare weight from the gross weight. The drained weight shall be reported to the nearest 0.1 ounce.

5/ The lot shall be rejected if the average drained weight of potato slices is less than 66.0 ounces.

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* 4.5.3.3 Salt content testing. Three filled and sealed tray pack cans shall be selected at random from the lot. The tray pack cans shall be individually tested for salt content in accordance with the Official Methods of Analysis of the Association of Official Analytical Chemists, except that preparation of the samples shall be as follows. The cans shall be opened and the entire contents of each can shall be separately blended in a Waring Blendor or equivalent. The test results shall be reported to the nearest 0.1 percent. Any result failing to conform to the salt requirement in 3.6 shall be classified as a major defect and shall be cause for rejection of the lot.

* 4.5.3.4 Commercial sterility testing. The sample size shall be one filled, sealed, and thermoprocessed tray pack can selected from each retort batch in the lot. The sample cans shall be tested for sterility by incubating the cans at 95 ± 5°F for 10 days. Any evidence of can swelling shall be classified as a critical defect and shall be cause for rejection of the lot.

* 4.5.4 Can condition examination. Examination of filled and sealed tray pack cans shall be in accordance with the United States Standards for Condition of Food Containers, except that inspection for labeling shall be as specified in 4.5.4.1. In addition, the following defect shall be classified as a major defect:

Evidence of buffing with an abrasive substance (see 5.1.1).

4.5.4.1 Can label examination. Labels shall be examined for defects in accordance with MIL-L-1497 (see 5.4), except that for self-adhering labels, the following additional defects shall apply:

Major: Label torn or scratched so as to obliterate any of the markings.

Minor: Air bubbles under label.
Label not properly adhered to can, for example, label raised or peeled back from edges or corners.

4.5.4.2 Label adhesive examination. When self-adhering labels are used, the adhesive shall be tested in accordance with ASTM D 3330.

4.5.5 Can closure examination. Can closures shall be examined visually and by teardowns in accordance with the can manufacturer's requirement and CFR 21, Part 113, Subpart D. Any nonconformance based on observation of can seam teardowns or of record of can seam teardowns shall be classified as a major defect and shall be cause for rejection of any involved product.

* 4.5.6 Vacuum examination. Cans shall be allowed to cool to 75 ± 5°F, held for at least 24 hours after sealing, and then examined for vacuum retention. To examine, lay a straight edge in the center of the lid along the length of the tray pack. Both ends of the straight edge shall touch the lid at the inside edge of the double seam. There shall be a visible gap between the straight edge and the lid for the entire distance of the label panel. Using a shorter

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straight edge, the same procedure shall be used across the width, in the center of the tray pack can. One measurement shall be made when examining a ribbed lid; lay the straight edge between the two center ribs along the length of the can. The inspection lot shall include only tray packs produced in a single shift on a single sealing machine. The sample size shall be 50 cans. Any nonconformance shall be classified as a major defect and shall be cause for rejection of the lot.

* 4.5.7 Test for rust spots on interior of tray pack cans. Where rust spots are suspected on interior of tray pack cans, the following test shall be performed:

a. Immerse a cotton swab in acetone or methyl ethyl ketone solvent and gently rub suspected spot. Handling of the solvent shall be in strict accordance with the guidelines of the manufacturer's Material Safety Data Sheet (MSDS).

b. If supected spot immediately disappears, it is not a rust spot.

c. If suspected spot remains, continue gently rubbing with the swab resoaked with solvent, and observe for disappearances or retention of spot. Replenish solvent as necessary.

d. If spot persists, and all but a very thin coating, or no coating remains, it will be scored as a rust spot. This can be confirmed further by rubbing with a finger. The rust spot will be felt as a slight bump over the substrate.

e. Any spot scored as a rust spot defect on interior of the tray pack can shall be cause for rejection of the lot.

4.5.8 Shipping container examination. Shipping containers shall be examined for defects in assembly, closure, and reinforcement (when applicable) in accordance with PPP-B-636. In addition, the following defects shall be classified as follows:

Major: National stock number, item description, contract number, or date of pack markings missing, incorrect, or illegible.
Reinforced with other than nonmetallic strapping or tape.
Dimensions of pads not as specified.
Interior packing with fiberboard liner or pads not as specified.

Minor: Other required markings missing, incorrect or illegible.
Arrangement or number of cans not as specified.

4.5.9 Unit load inspection. Inspection of unit loads shall be in accordance with the quality assurance provisions of MIL-L-35078.

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5. PACKAGING

* 5.1 Preservation. The product shall be preserved in accordance with level A.

* 5.1.1 Level A. One hundred and six ounces of food product shall be filled into a tray pack can conforming to MIL-C-44340 and sealed and thermoprocessed as specified in 3.4 and 3.5. The practice of reconditioning tray pack cans by buffing with an abrasive substance shall not be permitted (see 4.5.4). The interior coating of filled and sealed thermoprocessed cans shall be free of rust spots or scratches or fractures that penetrate through the coating when examined in accordance with 4.5.3.2.

5.2 Packing. The product shall be packed in accordance with level A, B, or C, as specified (see 6.1).

* 5.2.1 Level A packing. Four cans of product, preserved as specified in 5.1, shall be packed in a snug-fitting fiberboard box, constructed and closed in accordance with style RSC-L or HSC-L with an HSC full depth cover, grade V2s of PPP-B-636. The cans shall be packed flat, four in depth within the box, with the first two cans placed with the lids together and the next two cans with the lids together. The inside of each box shall be provided with a box liner and five fiberboard pads fabricated of grade V3c fiberboard. The height of the box liner shall be equal to the full inside depth of the box (+0 inch, -1/8 inch). Flute direction of the box liner shall be vertical. The pads shall be placed between the cans and on the top and bottom of the stacked cans. The pad dimensions shall be not less than 1/8 inch of the full length and width dimensions of the box. Each box shall be reinforced with nonmetallic strapping or pressure-sensitive adhesive filament-reinforced tape in accordance with the appendix of PPP-B-636. Shipping containers shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified (see 6.1), except that the unit load shall consist of 48 boxes with 12 boxes per course and four courses per load with all courses having the same pattern so as to create columnar stacking. When unit loads are strapped, strapping shall be limited to nonmetallic strapping, except for type II, class F loads.

5.2.2 Level B packing. Four cans of product, preserved as specified in 5.1, shall be packed as specified in 5.2.1 except the box shall be constructed of grade V3c, V3s, or V4s fiberboard.

* 5.2.3 Level C packing. Four cans of product, preserved as specified in 5.1, shall be packed in a snug-fitting fiberboard box, constructed and closed in accordance with style RSC-L, class domestic, grade 275 of PPP-B-636. The cans shall be packed flat, four in depth within the box with the first two cans placed with the lids together and the next two cans with the lids together. The inside of each box shall be provided with a box liner and five fiberboard pads. The height of the box liner shall be equal to the full inside depth of the box (+0 inch, -1/8 inch). Flute direction of the box liner shall be vertical. The pads shall be placed between the cans and on the top and bottom of the stacked cans. The pad dimensions shall be not less than 1/8 inch of the full length and

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width dimensions of the box and shall be fabricated of the same material as the box.

5.3 Unit loading. When specified (see 6.1), the product, packed as specified in 5.2.2 or 5.2.3, shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified, except that the unit load shall consist of 48 boxes with 12 boxes per course and four courses per load, with all courses having the same pattern so as to create columnar stacking. When unit loads are strapped, the strapping shall be limited to nonmetallic strapping, except for type II, class F loads.

5.4 Labeling. Each tray pack can shall be labeled in accordance with MIL-L-1497 and with the following:

- Official establishment number (for example, EST 38) or a three digit letter code identifying the establishment
- Lot number 1/
- Production shift number 1/
- Retort identification number 1/
- Retort cook number 1/

1/ The lot number shall be expressed as a four-digit Julian code. The first digit shall indicate the year of production and the next three digits shall indicate the day of the year (Example, March 19, 1990 would be coded as 0078). The Julian code shall represent the day the product was packaged and processed. Sub-lotting (when used) shall be represented by an alpha character immediately following the four-digit Julian code. Following the four-digit Julian code and the alpha character (when used), the other required code information shall be printed in the sequence as listed above.

In addition, the name of product shall be marked, stamping is permitted, on one 1001 by 200 side of the can. The labeling shall be legible when examined as specified in 4.5.4.1 after preparation of product in accordance with heating instructions. Paper labels are not permitted. Cans shall show the following statements:

TO HEAT IN WATER: Submerge unopened can in boiling water. Simmer gently for 40 to 45 minutes. Avoid overheating (can shows evidence of bulging).

CAUTION: Use care when opening as pressure may have been generated within the can.

TO HEAT IN OVEN: Either punch several holes in lid of can or open in usual manner leaving the loose lid in place. Place in a 350°F oven for approximately 35 to 40 minutes.

WARNING: Do not place unopened can in oven. This may cause the can to burst.

YIELD: Serves 18 portions of 2/3 cup each.

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As an alternative labeling method, a preprinted, self-adhering, 0.002-inch thick, clear polyester label printed with indelible black ink may be used. Self-adhering labels shall be applied after retorting. Pressure-sensitive adhesive shall require no preparation prior to application. Labels shall tack quickly and adhere without curling or breaking. The adhesive shall have a minimum adhesion of 60 ounces per inch width when examined as specified in 4.5.4.2. When self-adhering labels are used, the tray pack cans shall be labeled with the Julian code and a product code prior to retorting.

5.5 Marking.

5.5.1 Shipping containers. In addition to any special marking required by the contract or purchase order, shipping containers shall be marked in accordance with MIL-STD-129.

* 5.5.2 Unit loads. Unit loads shall be marked in accordance with MIL-L-35078. In addition, the following precautionary marking in capital letters larger than other markings shall be included:

CAUTION: DO NOT STACK PALLETS IN TRANSIT OR MORE THAN TWO HIGH IN STORAGE, UNLESS PALLET RACKS ARE USED.

6. NOTES

(This section contains information of a general or explanatory nature that maybe helpful, but is not mandatory.)

6.1 Ordering data. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
c. When a first article is required (see 3.1, 4.4, and 6.2).
d. Provisions for approved preproduction samples (see 3.6.1 and 6.2).
e. Level of packing required (see 5.2).
f. Type and class of unit load when unit loading is required (see 5.2.1 and 5.3).

6.2 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

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6.3 Appropriate level of pack. Based on the conditions known or expected to be encountered during shipment, handling, and storage of the specific item being procured, the procuring activity should select the appropriate level of pack in accordance with the criteria established in AR 700-15/NAVSUPINST 4030.28/AFR 71-6/MCO 4030.33A/DLAR 4145.7.

6.4 Subject term (key word) listing.

Canned foods
Combat field feeding
Food processing
Operational rations
Shelf stable

6.5 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:

Army - GL
Navy - SA
Air Force - 50

Review activities:

Army - MD, QM
Navy - MC
DP - SS

Preparing activity:

Army - GL

(Project 8940-0700)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-E-44231B	2. DOCUMENT DATE (YYMMDD) 1990 April 4

3. DOCUMENT TITLE
Escalloped Potatoes, Thermostabilized, Tray Pack

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME U.S. Army Natick RD&E Center	b. TELEPHONE (Include Area Code) (1) Commercial: 508-651-5221 (2) AUTOVON: 256-5221
c. ADDRESS (Include Zip Code) Commander, U.S. Army Natick RD&E Center ATTN: STRNC-ES Natick, MA 01760-5014	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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